

Labor Demand and Labor Supply Curves

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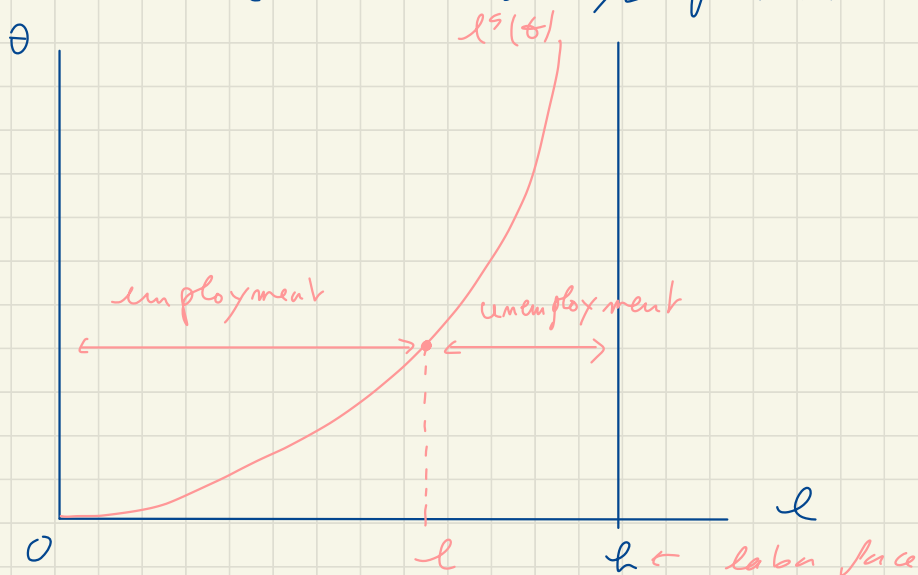
labor supply: # workers who find a job given labor-force participation & matching process.

$$l^s(\theta) = \hat{f}(\theta) \cdot h$$

job-finding proba

labor force

- $l^s(0) = 0$ b/c $\hat{f}(0) = 0$
- $\lim_{\theta \rightarrow \infty} l^s(\theta) = h$ b/c $\lim_{\theta \rightarrow \infty} \hat{f}(\theta) = 1$
- l^s is $\nearrow$ in θ b/c $\hat{f}$ is $\nearrow$ in θ
- l^s is concave in θ b/c $\hat{f}$ is concave in θ



labor demand # workers that firms want to
hire for given tightnesses α, θ and prices
 p, w [to maximize profits]

$$l^d(\alpha, \theta, p, w) = \left[\frac{f(\alpha) \alpha \alpha}{w/p} \right]^{1/(1-\alpha)} \left[\frac{1}{1 + \hat{\tau}(\theta)} \right]^{d/(1-\alpha)}$$

• $l^d(\alpha=0) = 0$ b/c $f(\alpha=0) = 0$
 • $l^d(\theta=\theta^m) = 0$ b/c $\hat{\tau}(\theta=\theta^m) = \infty$

• $\alpha \rightarrow \infty, f(\alpha) \rightarrow \infty$

• $\theta \rightarrow 0, \hat{\tau}(\theta) \rightarrow \hat{p}/(1-\hat{p}), 1 + \hat{\tau}(\theta) = 1/(1-\hat{p})$

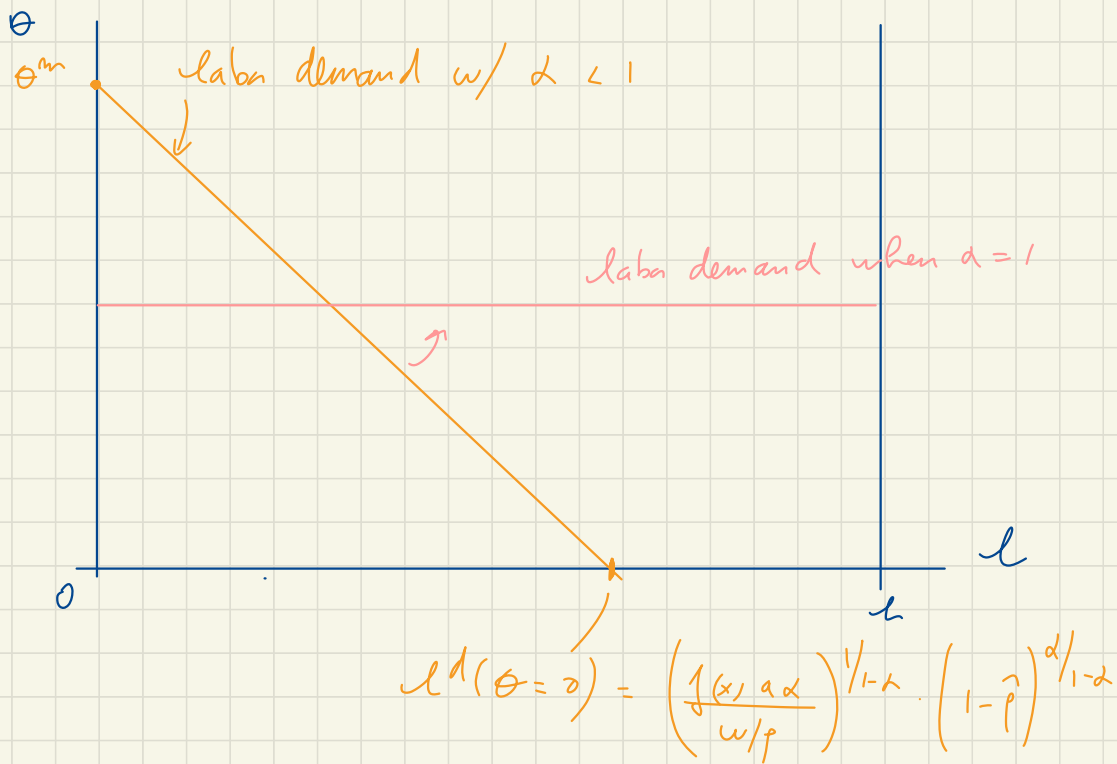
$$l^d(\theta=0) = \left[\frac{f(\alpha) \alpha \alpha}{w/p} \right]^{1/(1-\alpha)} [1 - \hat{p}]^{d/(1-\alpha)}$$

• l^d is $\uparrow$ in α (b/c $f(\alpha) \uparrow$ in α)

• l^d is $\downarrow$ in θ (b/c $\hat{\tau}(\theta) \uparrow$ in θ)

• l^d is $\downarrow$ in w/p

Labor-market diagram.



With linear production function.

$\alpha = 1$

Labor demand is

$$\underbrace{(l^d)^{1-\alpha}}_{\downarrow} = \left(\frac{f(x) \alpha \delta}{w/p} \right) \cdot \left(\frac{1}{1 + \hat{\tau}(\theta)} \right)^{\alpha} \quad \downarrow$$

Labor demand impl. 0, when $\alpha = 1$

$$1 + \hat{\tau}(\theta) = \frac{f(x) \alpha \delta}{w/p}$$

↳ no employment : degenerate demand

↳ $\theta^d(x, w, p) \rightarrow$ horizontal labor demand in (θ, e) diagram